

NATIONAL ECONOMICS UNIVERSITY
FACULTY OF DATA SCIENCE & AI

MIDTERM EXAMINATION (PAPER-BASED)

Course: Data Analysis with Python

Scope: Core Lectures 1 – 4 (Intro, NumPy, Pandas, Viz)

Course Code: DSAI1005 — **Duration:** 60 minutes

(No personal laptops, textbooks, or notes allowed)

Full Name: **Student ID:** **Exam Code:** 201-EN

Class / Cohort: **Exam Room:** **Seat No:**

MULTIPLE-CHOICE ANSWER SHEET (*Write letters A, B, C, or D in the corresponding boxes*):

Question	1	2	3	4	5	6	7	8	9	10
Answer										

Question	11	12	13	14	15	16	17	18	19	20
Answer										

SECTION I: MULTIPLE-CHOICE QUESTIONS (5.0 pts — 0.25 pts each)

- What is the core objective of the “Exploratory Data Analysis (EDA)” stage within the standardized 6-step data analysis lifecycle?
 - Immediately train complex black-box machine learning models to maximize benchmark accuracy.
 - Understand data structure, test hypotheses, uncover underlying patterns, and detect anomalies/outliers.
 - Optimize the physical storage engine and indexes in a relational database server.
 - Export raw tabular files to third-party vector illustration software for visual touch-ups.
- If every observation in a numerical dataset is multiplied by a constant factor of 3, the standard deviation of the new dataset will:
 - Increase by a factor of 9
 - Remain strictly unchanged
 - Increase by a factor of 3
 - Equal the square root of the original value
- In a standard Tukey Box Plot, an observation x is formally classified as a potential outlier if it falls outside which of the following intervals (where $IQR = Q_3 - Q_1$)?
 - $[Q_1 - 1.5 \times IQR, Q_3 + 1.5 \times IQR]$
 - $[Q_1 - 3.0 \times IQR, Q_3 + 3.0 \times IQR]$
 - $[\text{Mean} - 1.5 \times \text{Std}, \text{Mean} + 1.5 \times \text{Std}]$
 - $[0, Q_3 + IQR]$
- The Pearson correlation coefficient between two continuous variables X and Y evaluates to $r = 0$. Which of the following statements is most statistically sound?
 - X and Y are completely independent and share no mathematical relationship whatsoever.
 - There is no linear association between X and Y , although a strong non-linear relationship may still exist.
 - Whenever X increases by 1 unit, Y remains fixed at 0.
 - Both variables X and Y must follow a standard normal distribution.

5. What array is generated by the NumPy expression `np.arange(3, 15, 4)`?

- A. [3, 7, 11, 15] B. [3, 7, 11] C. [4, 8, 12] D. [3, 6, 9, 12]

6. Consider the following NumPy snippet:

```
import numpy as np
m = np.array([[1, 2, 3], [4, 5, 6], [7, 8, 9]])
print(m[::-1, 1])
```

What output is printed to the console?

- A. [2, 5, 8] B. [8, 5, 2] C. [9, 6, 3] D. [7, 4, 1]

7. Array *A* has shape (5, 1, 3) and array *B* has shape (4, 1). Under NumPy broadcasting rules, what is the shape of the resulting array from `A * B`?

- A. (5, 4, 3) B. (5, 1, 3) C. (4, 5, 3) D. Raises
ValueError

8. Which NumPy function performs element-wise conditional substitution, returning elements chosen from *X* or *Y* depending on a boolean condition (equivalent to a vectorized ternary operator)?

- A. `np.select()` B. `np.where()` C. `np.filter()` D. `np.choose()`

9. For a two-dimensional array `arr` with shape (4, 5), what is the shape of the result returned by `np.sum(arr, axis=0)`?

- A. (4,) B. (5,) C. (1,) D. (4, 5)

10. Which primary Pandas data structure represents a one-dimensional labeled array capable of holding any data type?

- A. DataFrame B. Series C. IndexList D. NDArray

11. Given the Pandas Series `s = pd.Series([10, 20, 30, 40], index=['a', 'b', 'c', 'd'])`, what elements are returned by the slice `s.loc['a':'c']`?

- A. Only elements with index 'a' and 'b' sliced
B. All three elements with index 'a', 'b', and 'c'
C. A syntax error because string labels cannot be D. Only the single element with index 'c'

12. In Pandas, which method is specifically designed to impute or replace missing entries (NaN) with a designated value or summary statistic (e.g., column mean)?

- A. `df.dropna()` B. `df.fillna()` C. `df.replace_na()` D. `df.impute()`

13. Which statement correctly filters records in DataFrame `df` where `credit_score >= 700` OR `income > 50000000`?
- A. `df[(df['credit_score'] >= 700) or (df['income'] > 50000000)]`
 - B. `df[(df['credit_score'] >= 700) | (df['income'] > 50000000)]`
 - C. `df[(df['credit_score'] >= 700) || (df['income'] > 50000000)]`
 - D. `df.filter(credit_score >= 700 or income > 50000000)`

14. Given the following Pandas snippet:

```
import pandas as pd
df = pd.DataFrame({'Dept': ['Sales', 'IT', 'IT', 'Sales'], 'Salary': [30, 50, 70, 40]})
print(df.groupby('Dept')['Salary'].agg(['min', 'max']).loc['IT', 'max'])
```

What value is printed?

- A. 30
 - B. 50
 - C. 70
 - D. 60
15. When executing an inner merge between two DataFrames using `pd.merge(df1, df2, on='id', how='inner')`, a row appears in the output DataFrame if and only if:
- A. The key 'id' is present in either `df1` or `df2`.
 - B. The key 'id' matches simultaneously in both `df1` and `df2`.
 - C. `df1` contains strictly more rows than `df2`.
 - D. The key column contains strictly unique non-null integers.
16. How many distinct subplot axes (Axes) are created by the call `fig, axes = plt.subplots(2, 3)` in Matplotlib?
- A. 5 axes
 - B. 6 axes
 - C. 2 axes
 - D. 3 axes
17. Which Seaborn function is best suited to estimate and visualize a smooth, continuous probability density curve (Kernel Density Estimation) for a financial variable?
- A. `sns.kdeplot()`
 - B. `sns.scatterplot()`
 - C. `sns.countplot()`
 - D. `sns.barplot()`
18. In Seaborn, which high-level function automatically renders an $n \times n$ grid showing pairwise scatter plots across all quantitative features and univariate distributions along the diagonal?
- A. `sns.jointplot()`
 - B. `sns.pairplot()`
 - C. `sns.heatmap()`
 - D. `sns.catplot()`
19. In Matplotlib, which utility function is commonly called prior to display or export to automatically adjust subplot margins, preventing axes labels and titles from overlapping?
- A. `plt.align_axis()`
 - B. `plt.tight_layout()`
 - C. `plt.clean_view()`
 - D. `plt.auto_padding()`
20. To compare the mean values of a numerical metric across categorical groups while displaying confidence intervals (error bars) by default, which Seaborn function is optimal?
- A. `sns.lineplot()`
 - B. `sns.barplot()`
 - C. `sns.histplot()`
 - D. `sns.heatmap()`

SECTION II: SHORT-ANSWER & CODE COMPREHENSION (5.0 pts)

Question 1: Vectorized Computations & Multidimensional Arrays with NumPy (1.0 pt) — [Week 2]

Consider the exam score matrix of 4 students across 3 subjects (Math, Probability, Programming) and the corresponding course credit weights:

```
import numpy as np

scores = np.array([
    [8.0, 7.0, 9.0],
    [6.0, 8.0, 7.0],
    [9.0, 9.0, 8.5],
    [5.0, 6.0, 6.5]
]) # Shape: (4, 3)

weights = np.array([0.2, 0.3, 0.5]) # Shape: (3,)
```

Tasks:

- (0.5 pts)** Name the fundamental NumPy mechanism that enables direct multiplication between `scores` of shape `(4, 3)` and `weights` of shape `(3,)`. Write the exact numerical matrix for `weighted_scores = scores * weights`.
- (0.5 pts)** Write a single NumPy statement to calculate the 1D array `gpa` (representing the weighted grade point average for each student by summing along the row axis of `weighted_scores`), and determine the exact output printed by: `print(scores[gpa >= 8.0, 0])`

Work Area — Question 1:

a) Name of mechanism:

Resulting matrix `weighted_scores`:

b) NumPy statement to compute `gpa`:

`gpa = .....`

Exact output printed by the print statement:

(Brief reasoning if applicable):

Question 2: Data Cleaning & Aggregation with Pandas (1.5 pts) — [Week 3]

You are given a Pandas DataFrame `df_trans` representing retail banking transaction logs with columns: 'branch' (branch code), 'amount' (transaction value in VND), 'fee' (service charge), and 'status' (transaction outcome: 'SUCCESS' or 'FAILED'). The column 'fee' contains several missing entries (NaN).

Tasks (write correct and concise Pandas code):

- (0.5 pts)** Write Pandas code to filter rows satisfying two conditions: 'status' equals 'SUCCESS' and 'amount' $\geq 5\,000\,000$ VND. Simultaneously, impute all missing values (NaN) in the 'fee' column of this filtered subset with 0.
- (1.0 pts)** Using the filtered dataset from part (a), construct a chained Pandas expression (using `groupby` and `agg`) to group by branch ('branch'), compute: total transaction volume named 'total_amount', average fee named 'avg_fee', and transaction count named 'trans_count'. Sort the final summary table in descending order of 'total_amount'.

*Work Area — Question 2:**a) Filtering rows and imputing missing values (NaN):*

`df_filtered = .....`

`df_filtered['fee'] = .....`

b) Groupby, aggregation, and sorting expression:

`df_branch_summary = (.....)`

.....

.....

.....

.....)

Question 3: Multi-plot Layouts with Matplotlib (1.0 pt) — [Week 4]

Given a DataFrame `df_kpi` tracking monthly performance with columns 'month' (integers 1 to 12) and 'profit' (net profit in million VND).

Tasks (complete the code template by filling in the blanks):

- (0.5 pts)** Write the statement to initialize a figure of size 12×4 inches containing 1 row and 2 columns of subplots, storing the objects in `fig` and array `axes`.
- (0.5 pts)** On `axes[0]`, render a line plot (`plot`) illustrating monthly profit trajectory (blue line with circular markers 'o'). On `axes[1]`, render a bar chart (`bar`) of monthly profit. Finally, apply layout optimization.

Work Area — Question 3:

```
import matplotlib.pyplot as plt
```

```
fig, axes = .....
axes[0].plot( ..... )
axes[1].bar( ..... )
(# Command to prevent overlapping labels/titles)
```

Question 4: Exploratory Visualization & Business Insights (EDA) (1.5 pts) — [Weeks 1 & 4]

A bank's Risk Analytics division evaluates transaction data to detect card fraud. The DataFrame `df_cards` contains: 'tx_amount' (monetary value), 'card_type' (categorical: 'Credit' or 'Debit'), and fraud flag 'is_fraud' (1 for fraudulent, 0 for legitimate).

Tasks:

- (0.75 pts)** Choose the most appropriate statistical chart type to inspect the distribution and detect potential extreme outliers of 'tx_amount' across fraud labels, stratified by card type. Write a single line of **Seaborn** (`sns`) code to render this visualization.
- (0.75 pts)** The resulting plot reveals: For 'Debit' cards, both legitimate and fraudulent transactions cluster at low values with no significant outliers. Conversely, for 'Credit' cards, fraudulent transactions display an extreme outlier spread from 50 to 200 million VND, whereas legitimate ones remain under 10 million VND. *Formulate 2 concise analytical takeaways / policy recommendations (under 50 words) to guide the bank's automated fraud prevention rule.*

Work Area — Question 4:

a) Chosen chart type:

Seaborn statement:

b) Analytical Takeaways & Policy Recommendations:

- Takeaway 1:

- Policy 2:

————— **END OF EXAM** —————

(Proctors will not answer questions regarding exam content. Hand in all exam sheets upon completion.)